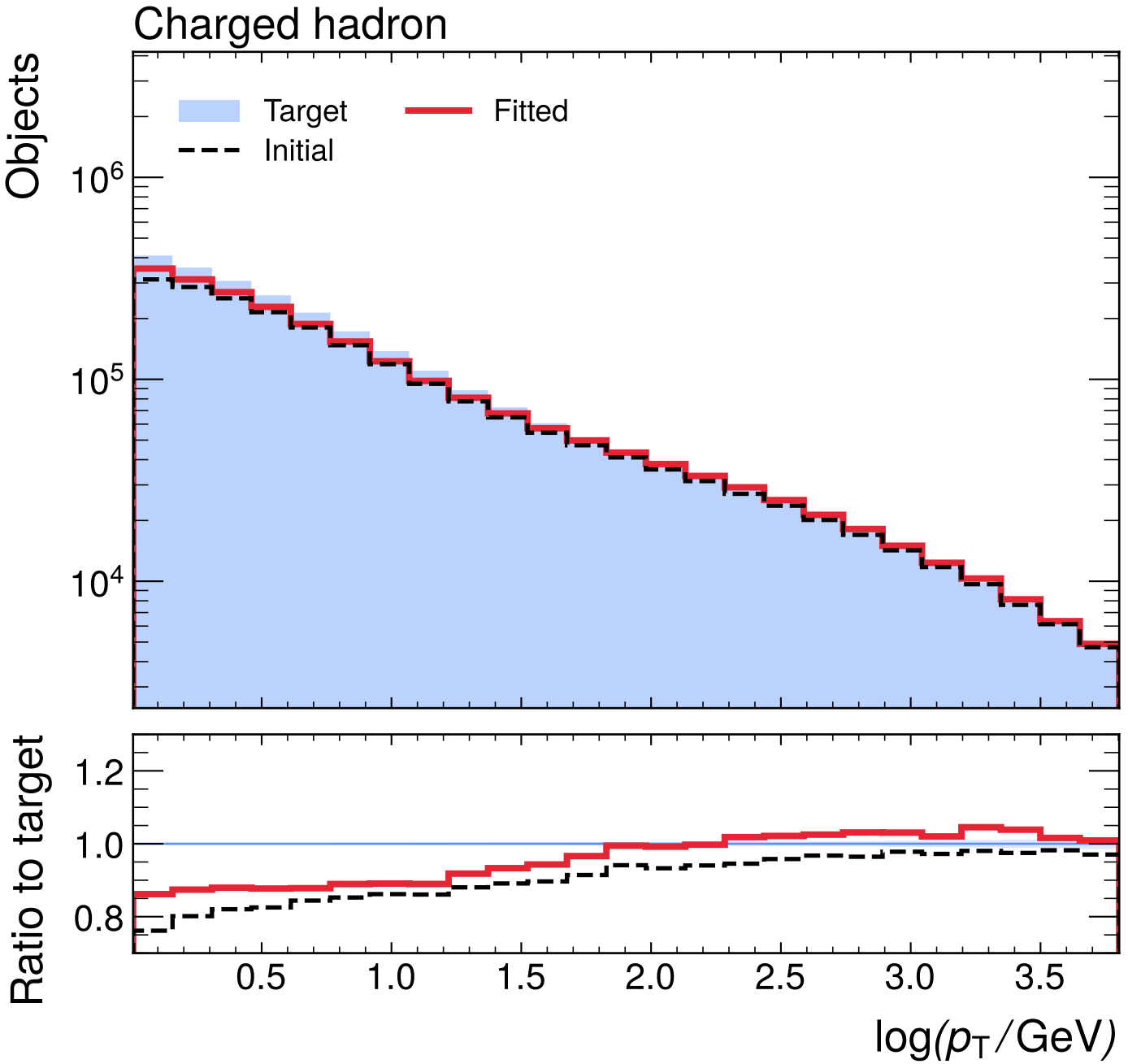
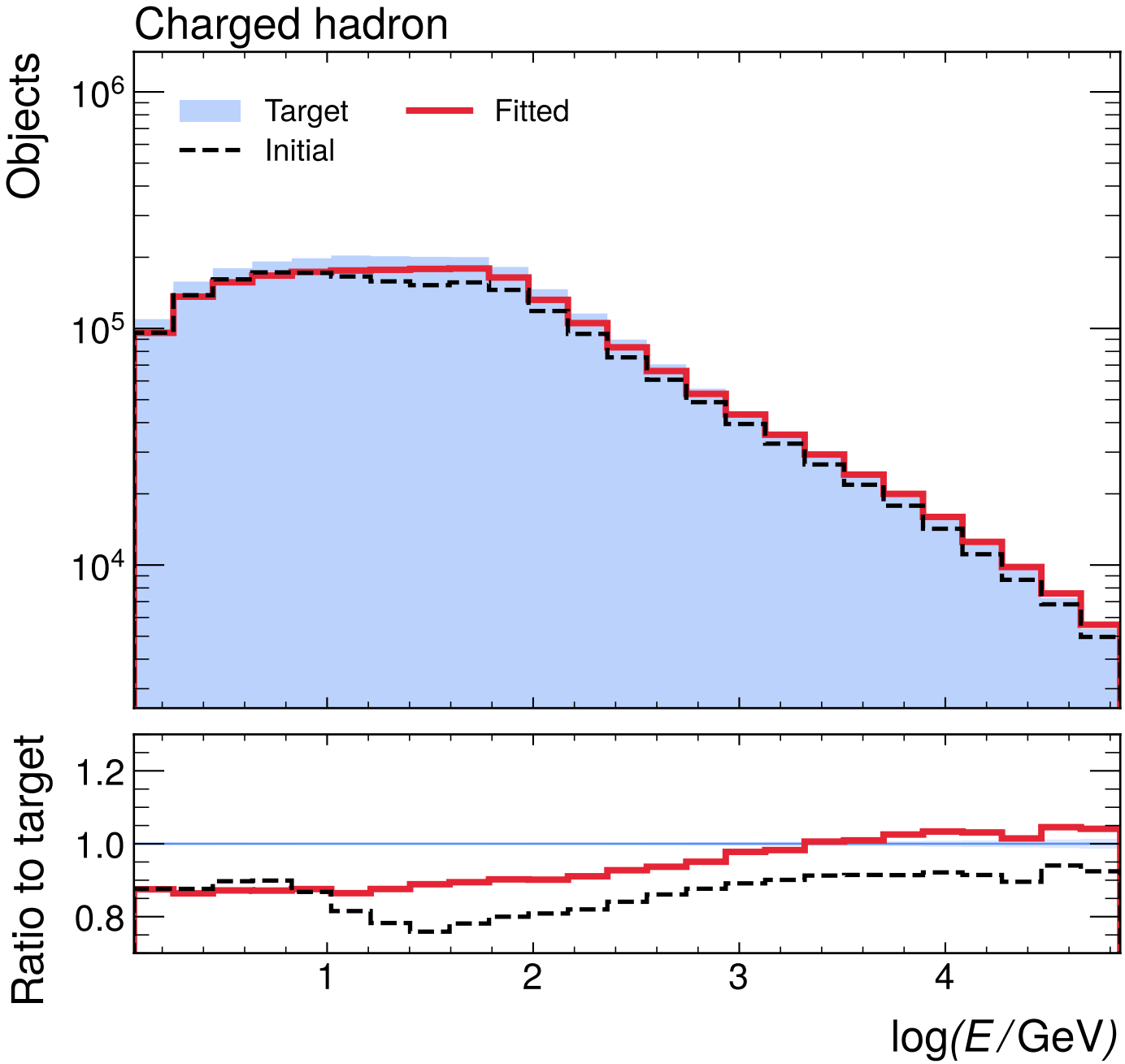
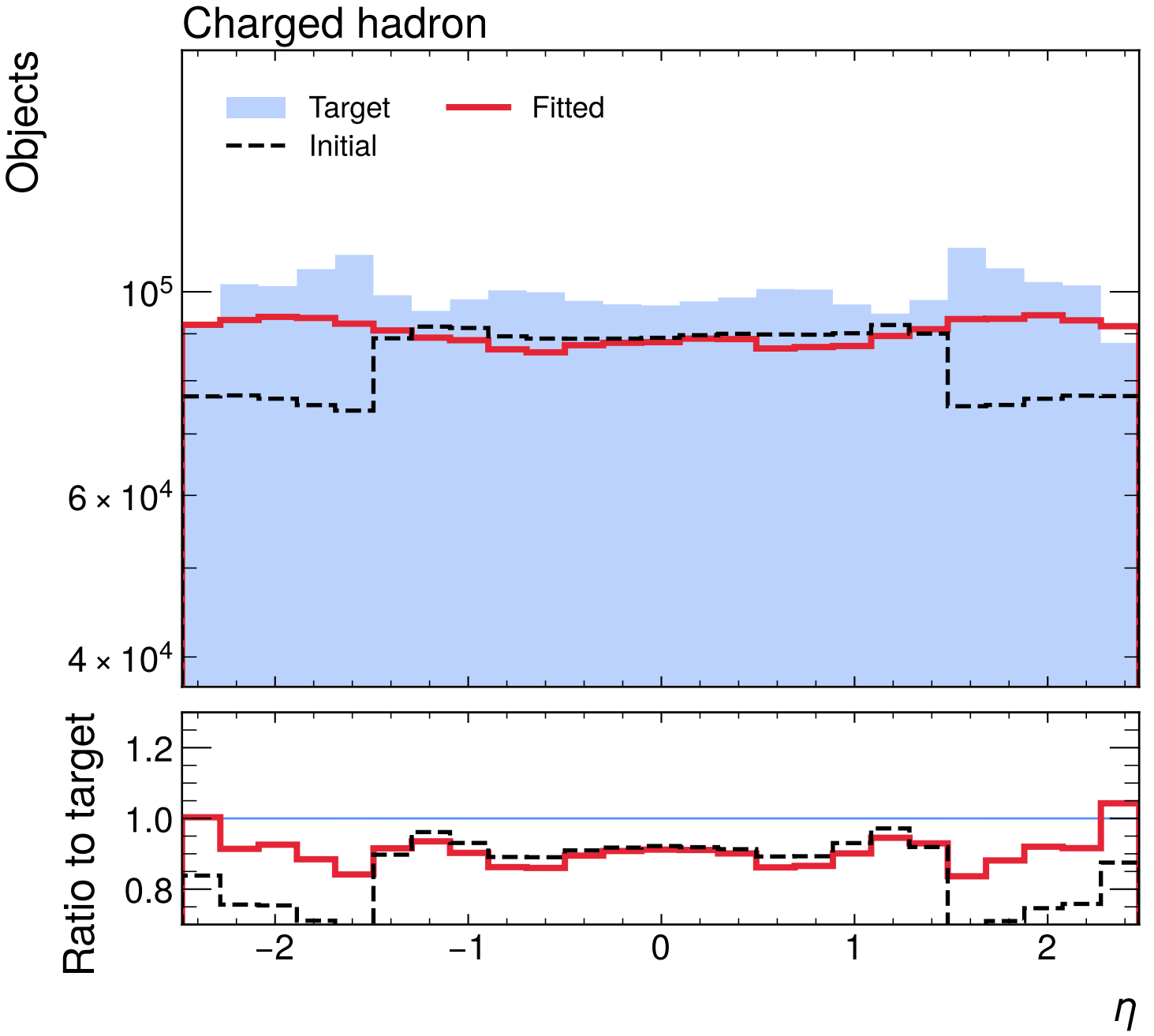
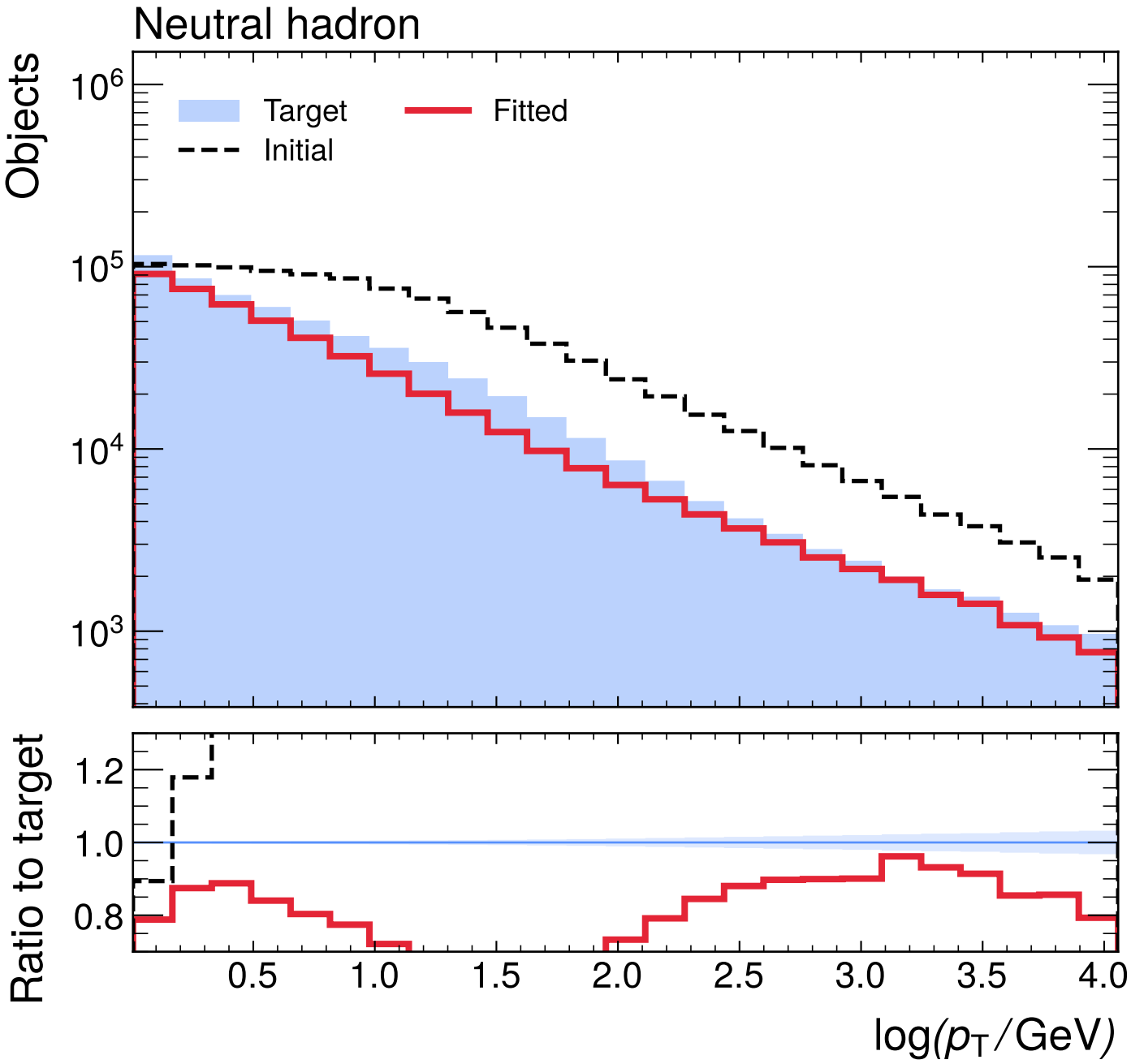
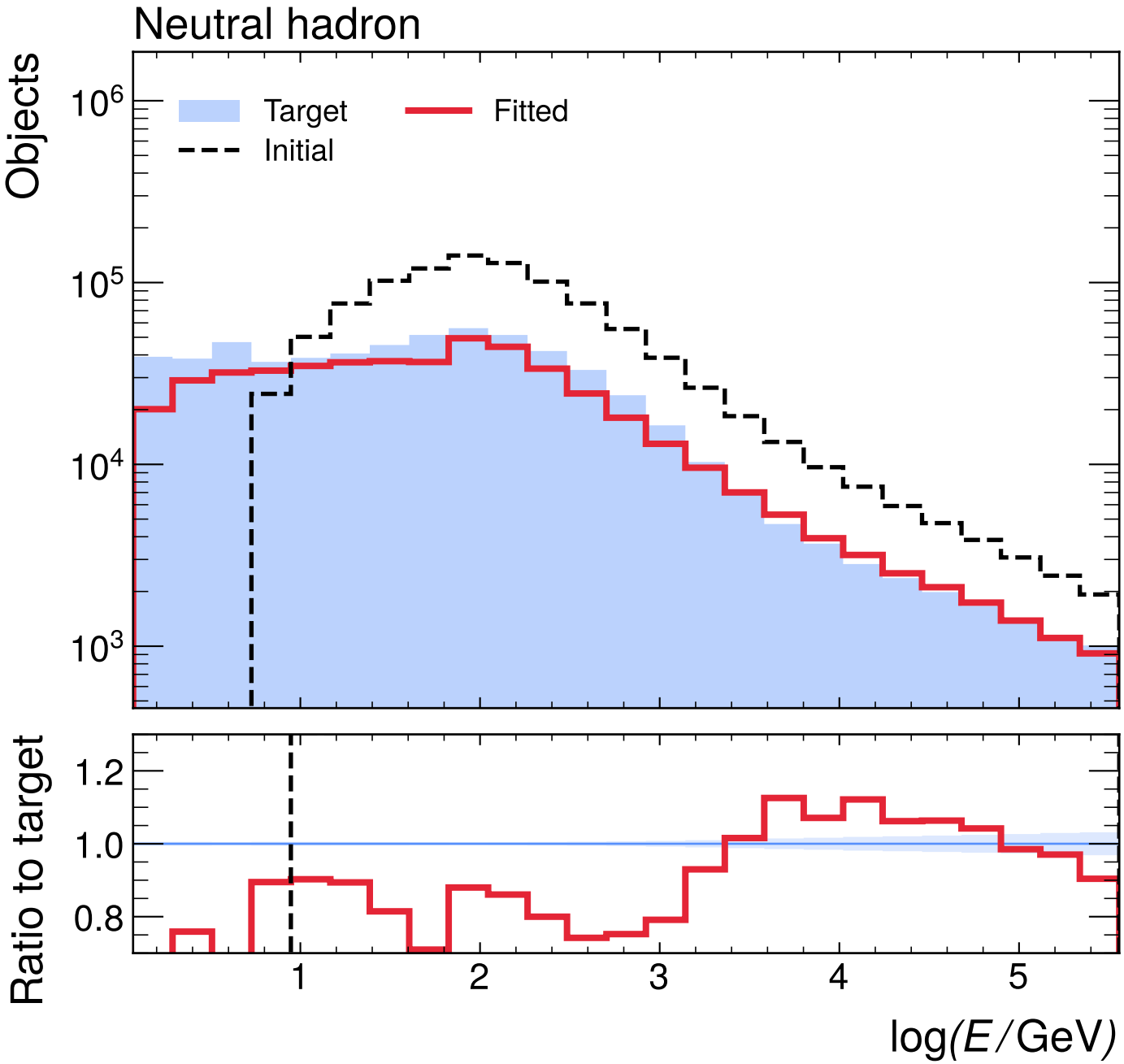
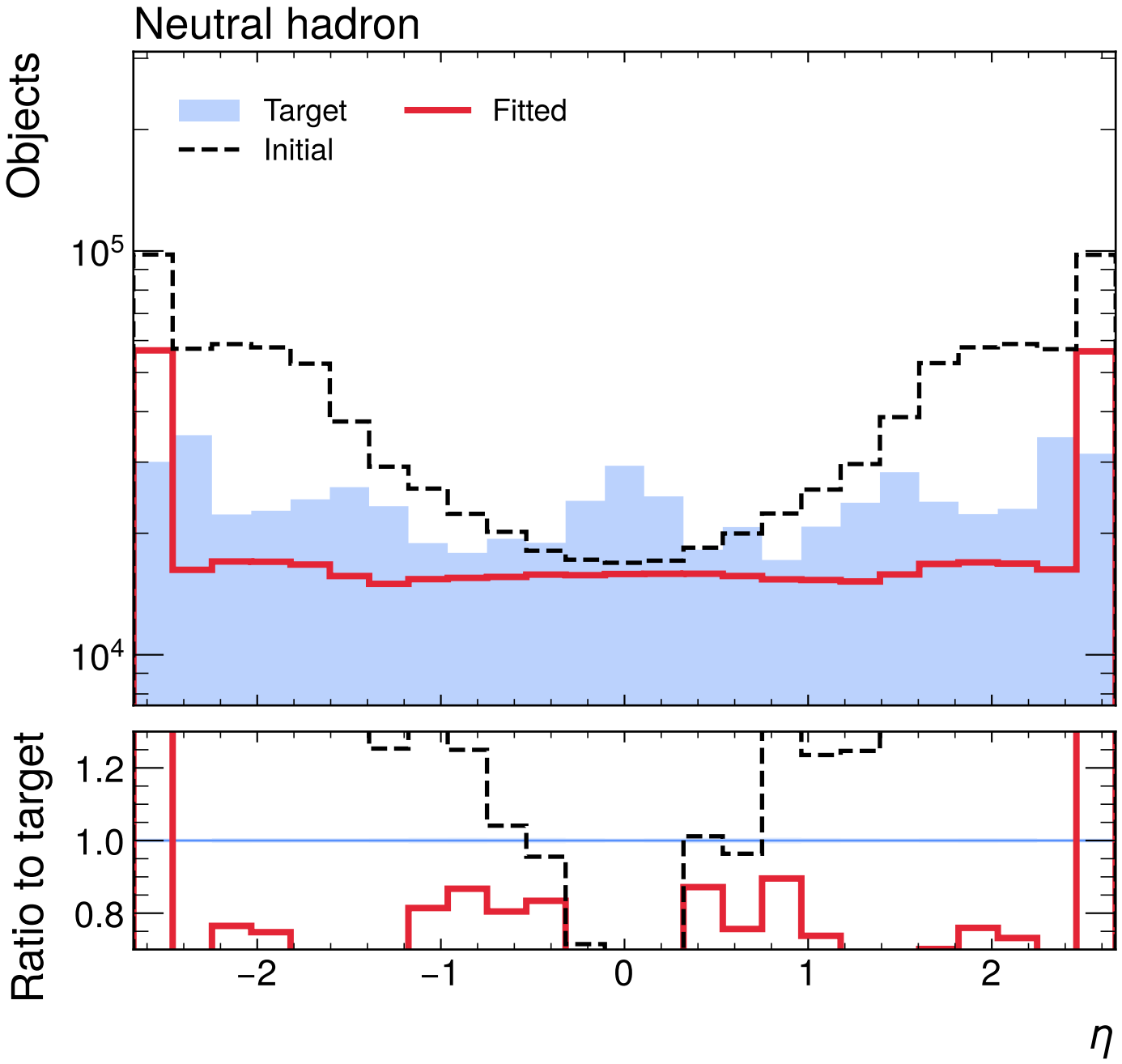

Charged hadron
Objects
Target
Initial
Fitted
10^6
10^5
10^4
Ratio to target
1.2
1.0
0.8
0.5
1.0
1.5
2.0
2.5
3.0
3.5
$\log(p_T/\mathrm{GeV})$

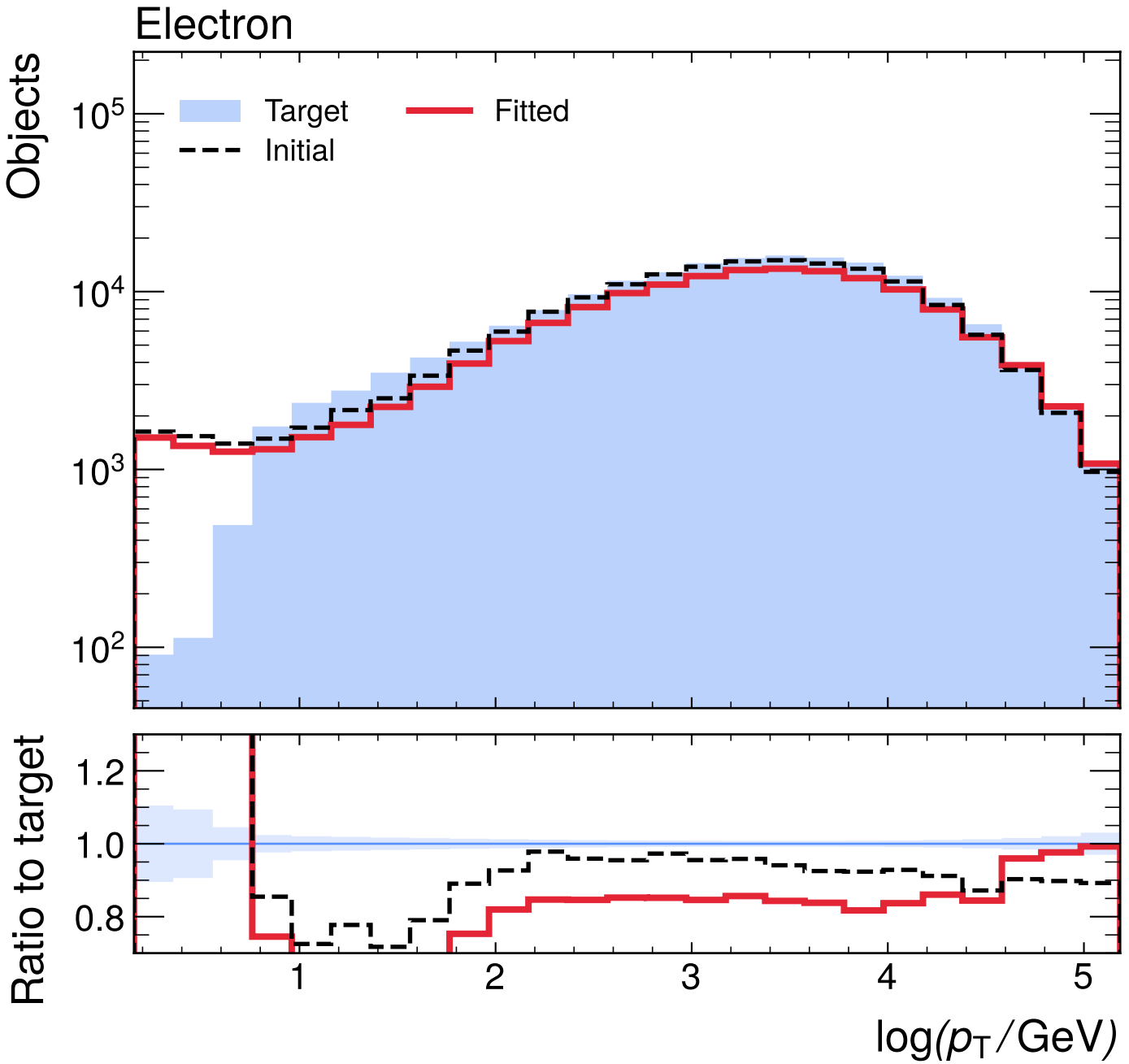

Charged hadron
Objects
Target
Initial
Fitted
10^6
10^5
10^4
Ratio to target
1.2
1.0
0.8
1
2
3
4
log(E/GeV)

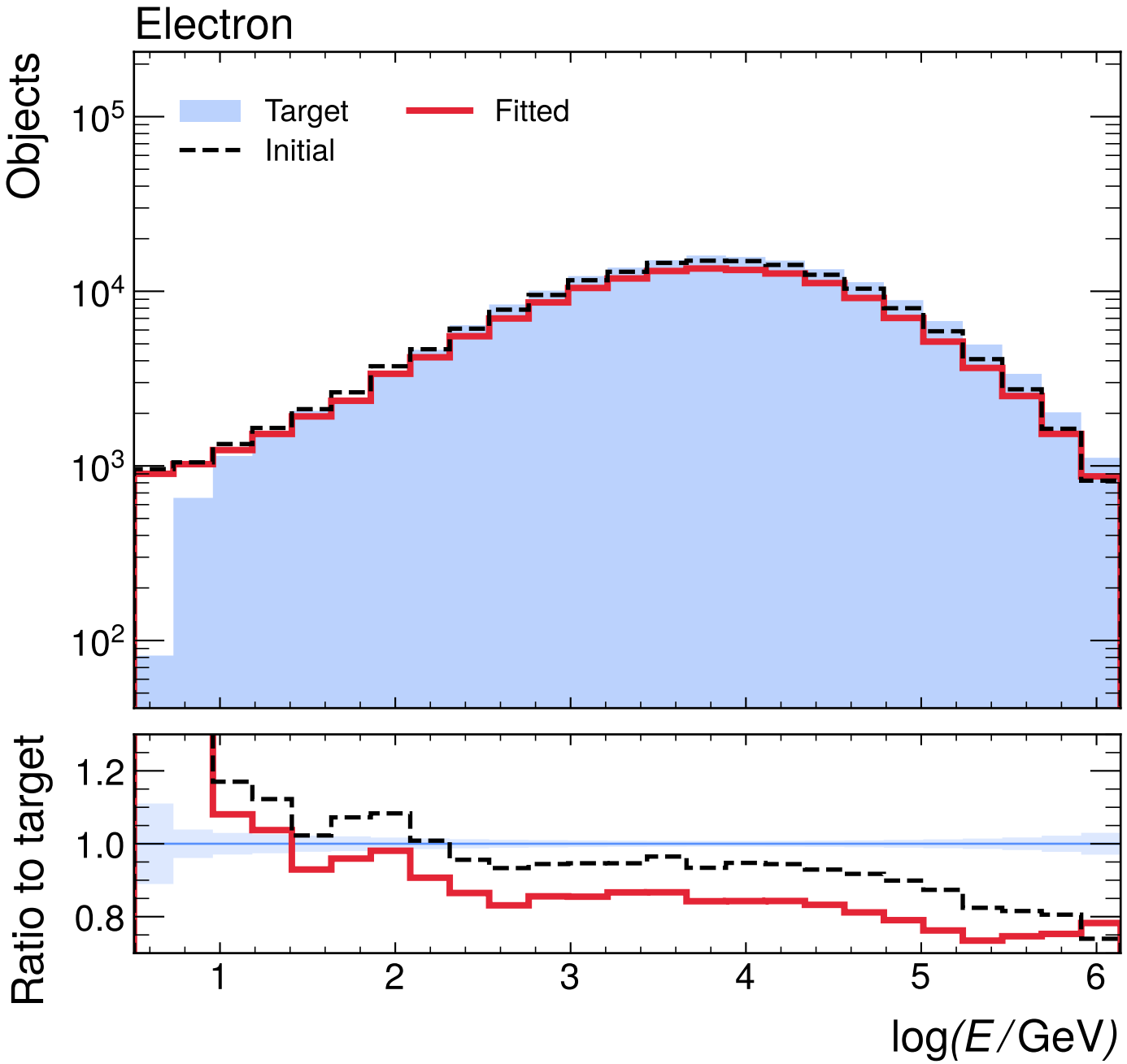

Charged hadron
Objects
Target
Fitted
Initial
10^5
6×10^4
4×10^4
Ratio to target
1.2
1.0
0.8
−2
−1
0
1
2
η

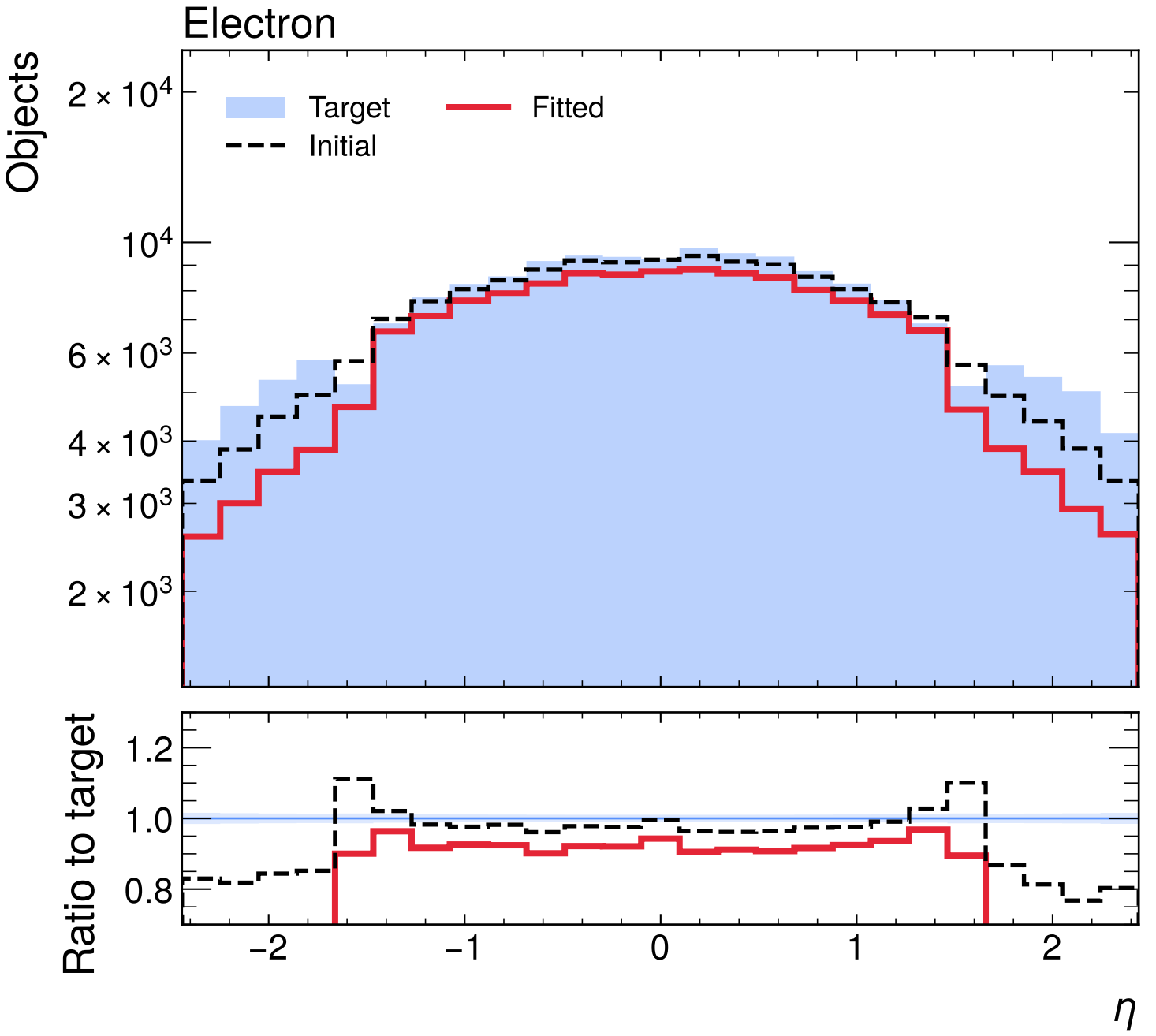

Neutral hadron
Objects
Target
Initial
Fitted
10^6
10^5
10^4
10^3
Ratio to target
1.2
1.0
0.8
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
$\log(p_T/\mathrm{GeV})$

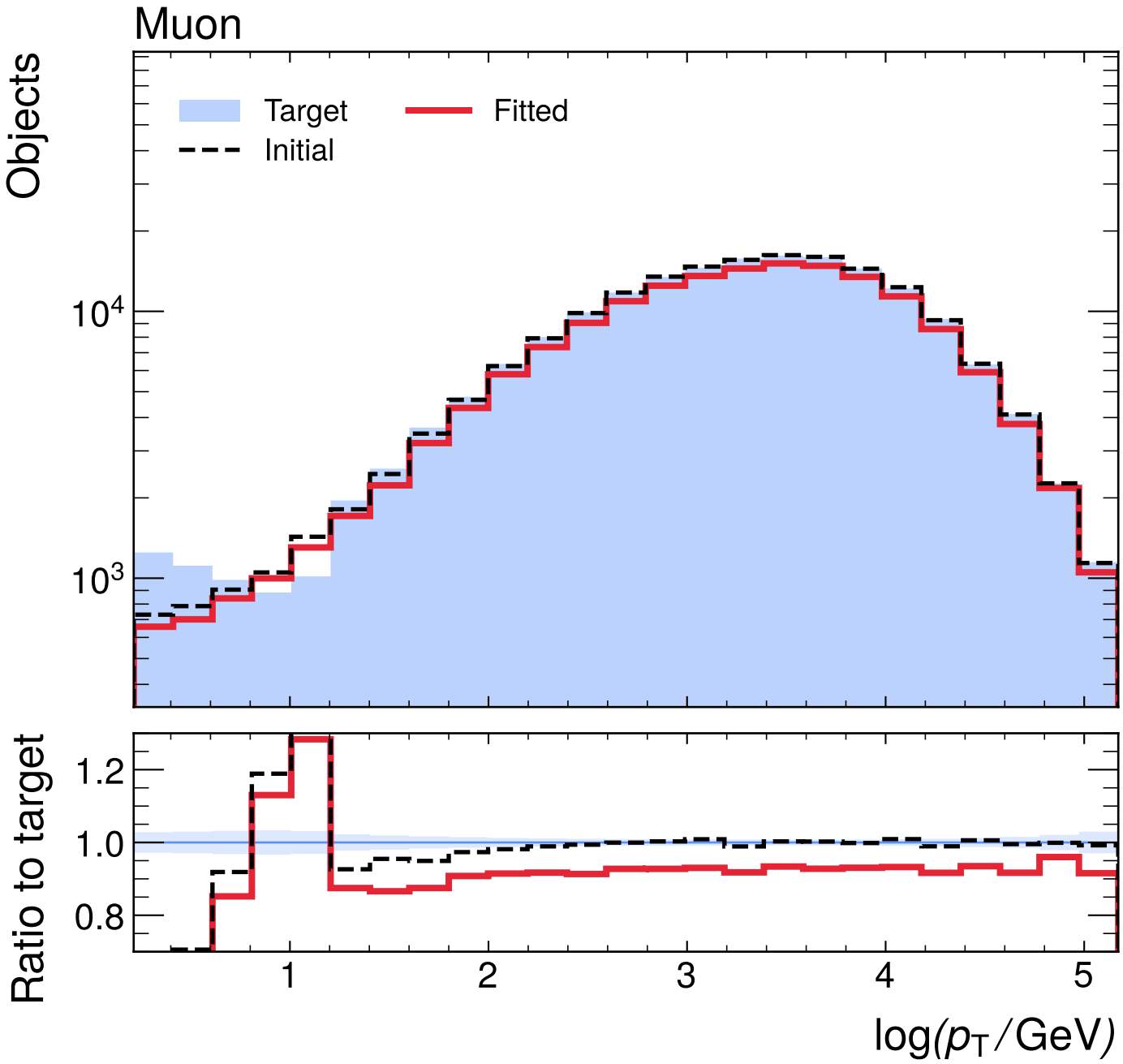

Neutral hadron
Objects
Target
Initial
Fitted
10^6
10^5
10^4
10^3
Ratio to target
1.2
1.0
0.8
1
2
3
4
5
log(E/GeV)

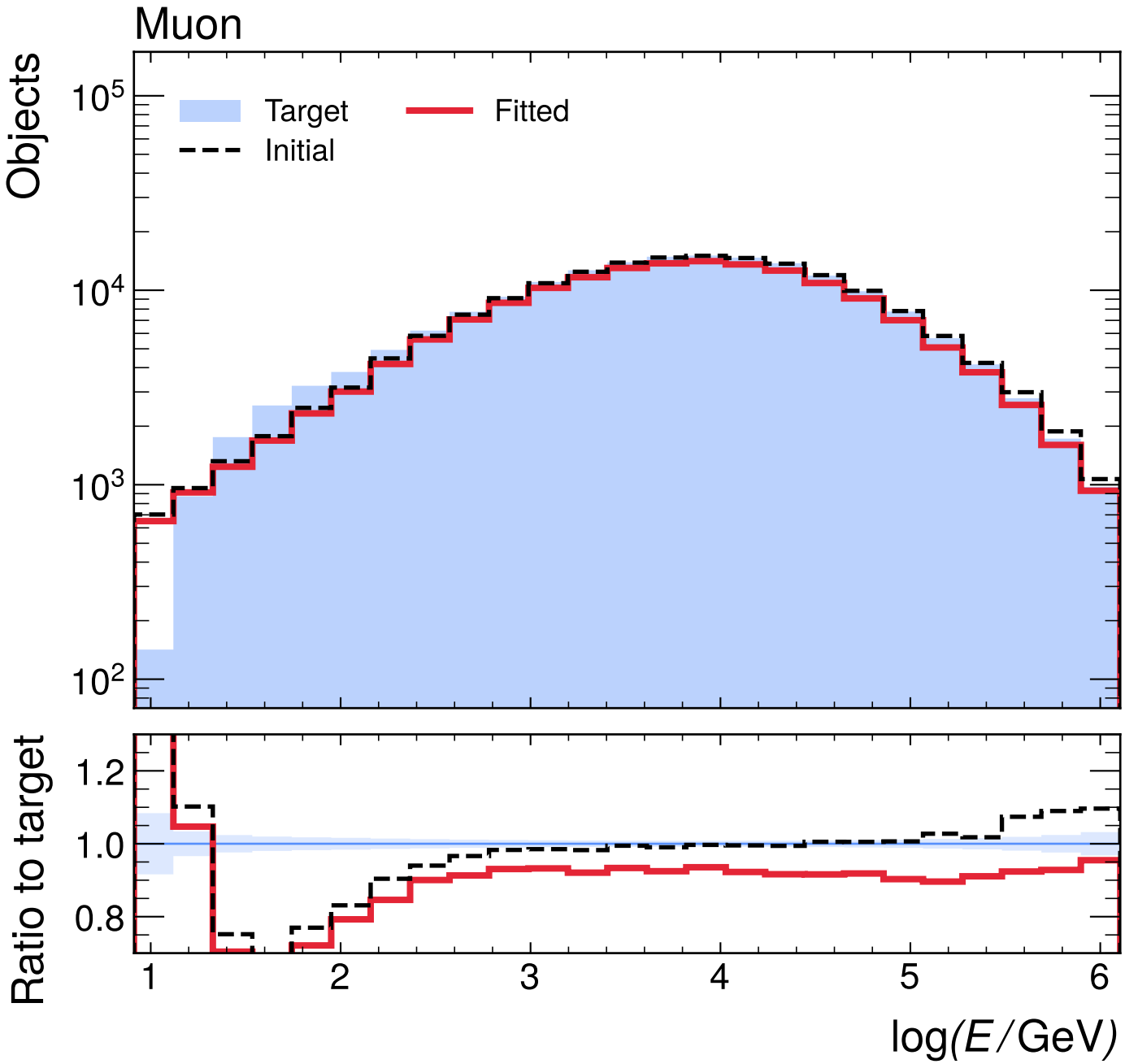

Neutral hadron
Objects
Target
Fitted
Initial
10^5
10^4
Ratio to target
1.2
1.0
0.8
−2
−1
0
1
2
η

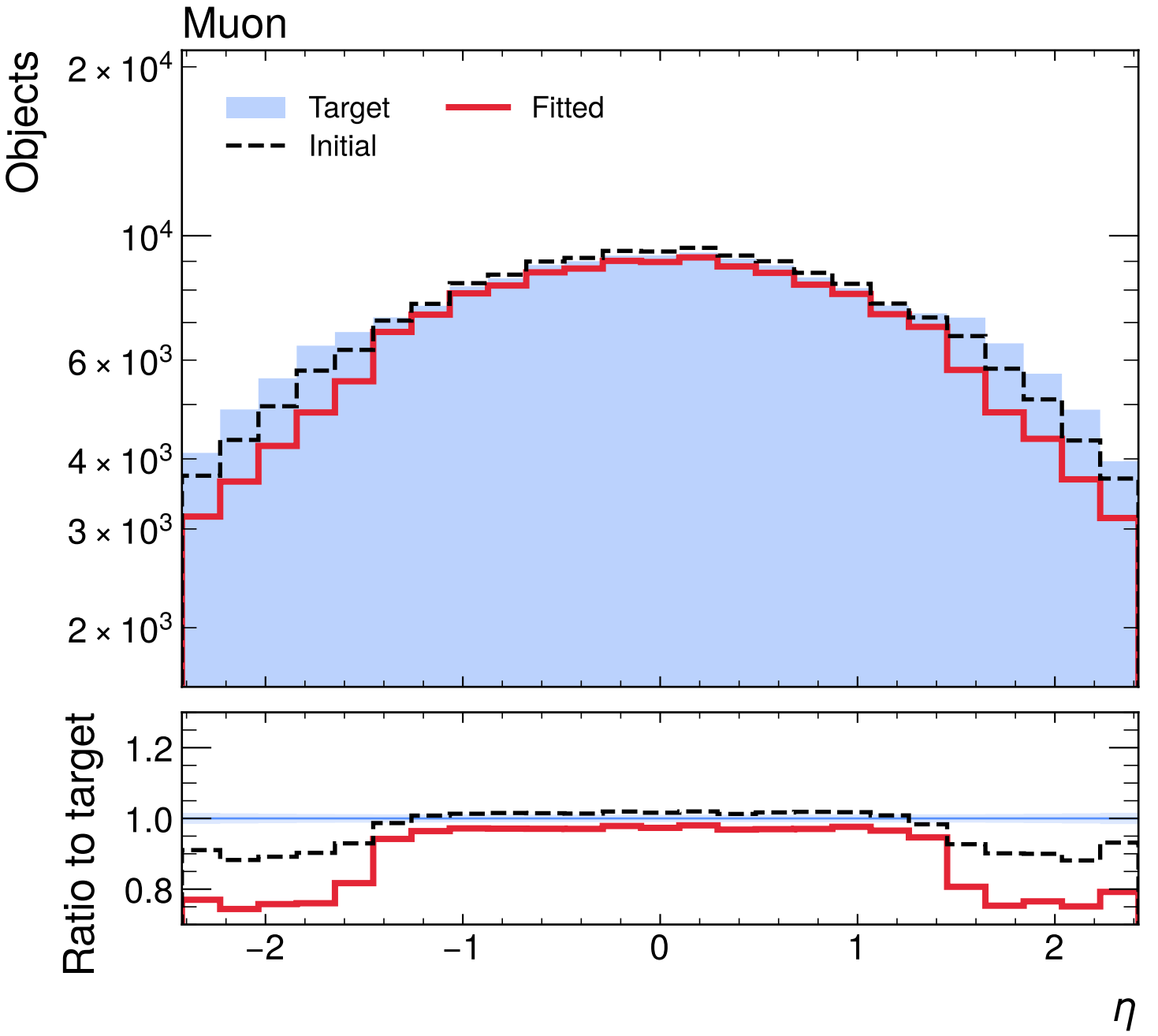

Electron
Objects
Target
Initial
Fitted
10^5
10^4
10^3
10^2
Ratio to target
1.2
1.0
0.8
1
2
3
4
5
log(p_T/GeV)

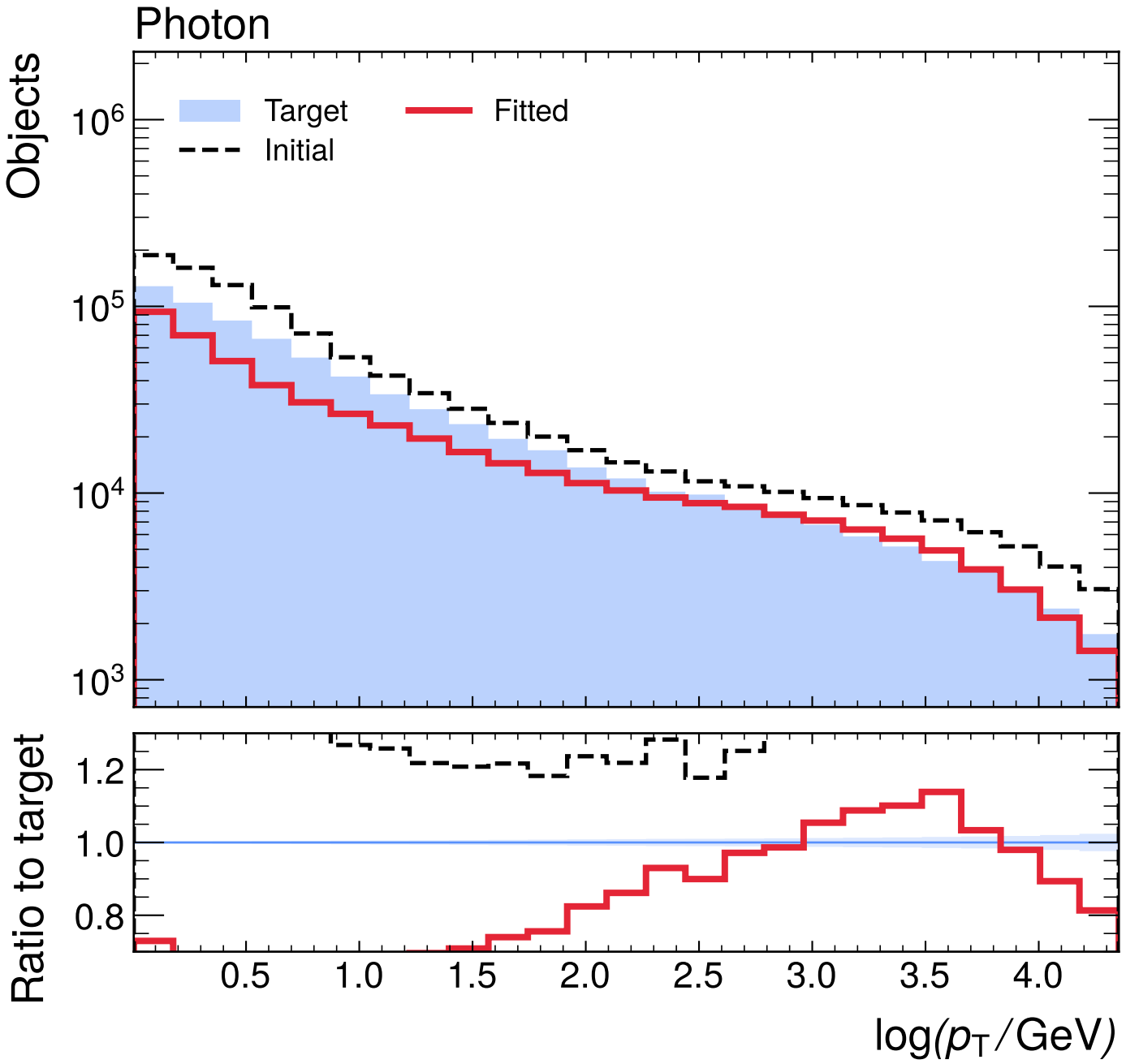

Electron
Objects
Target
Initial
Fitted
10^5
10^4
10^3
10^2
Ratio to target
1.2
1.0
0.8
1
2
3
4
5
6
log(E/GeV)

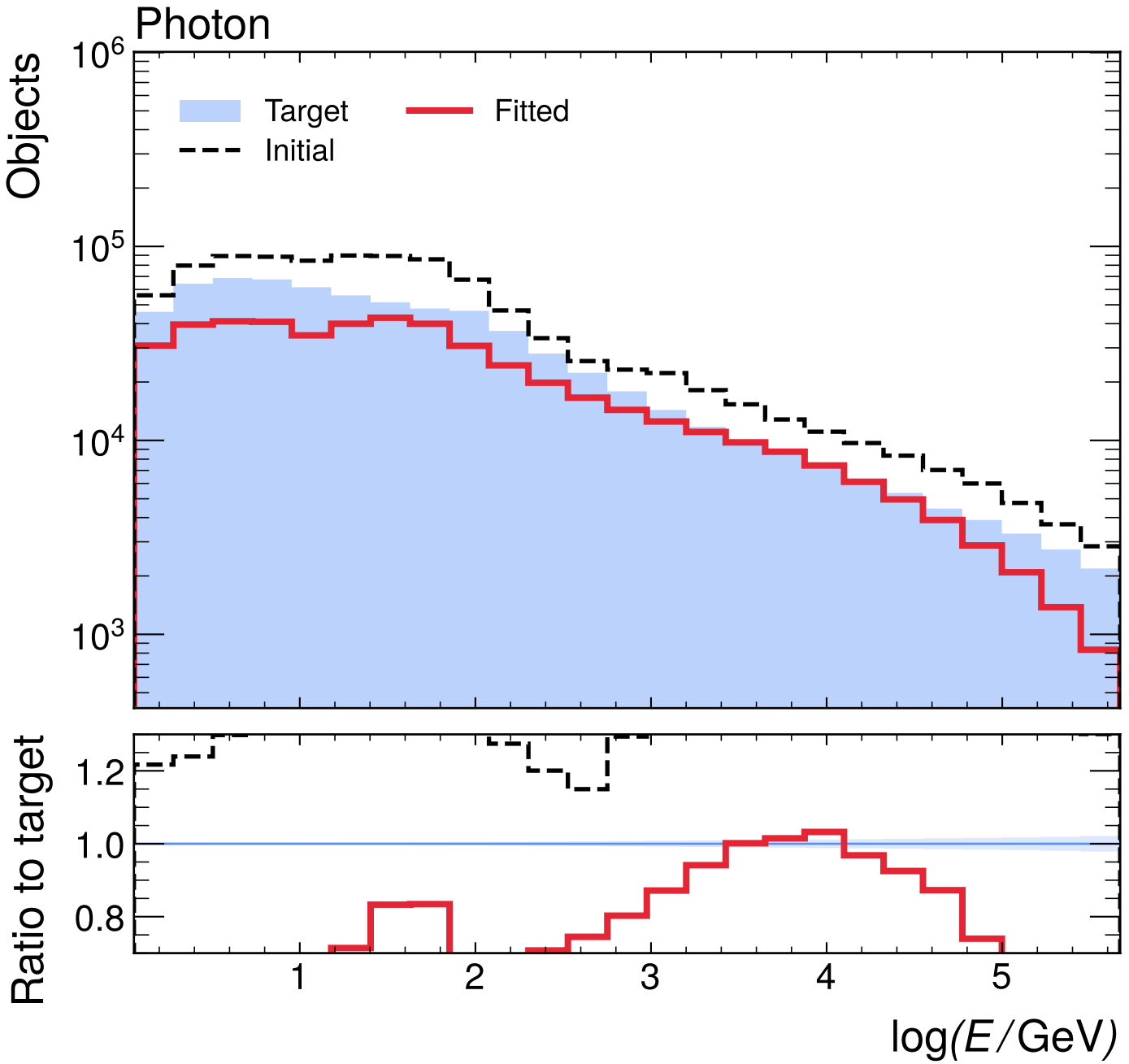

Electron
Objects
Target
Initial
Fitted
2×10^4
10^4
6×10^3
4×10^3
3×10^3
2×10^3
Ratio to target
1.2
1.0
0.8
−2
−1
0
1
2
η

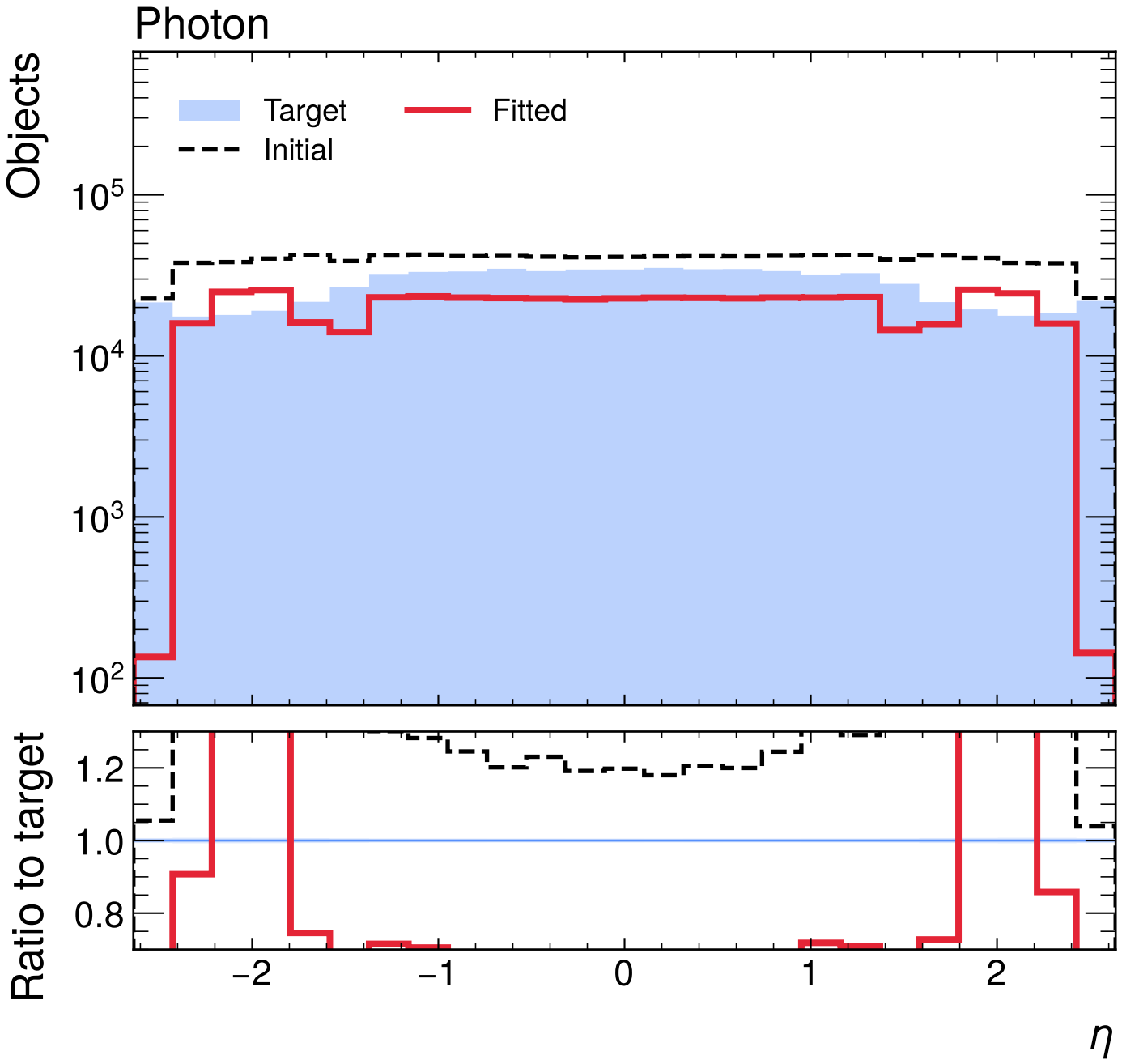

Muon
Objects
Target
Initial
Fitted
10^4
10^3
Ratio to target
1.2
1.0
0.8
1
2
3
4
5
$\log(p_T/\mathrm{GeV})$

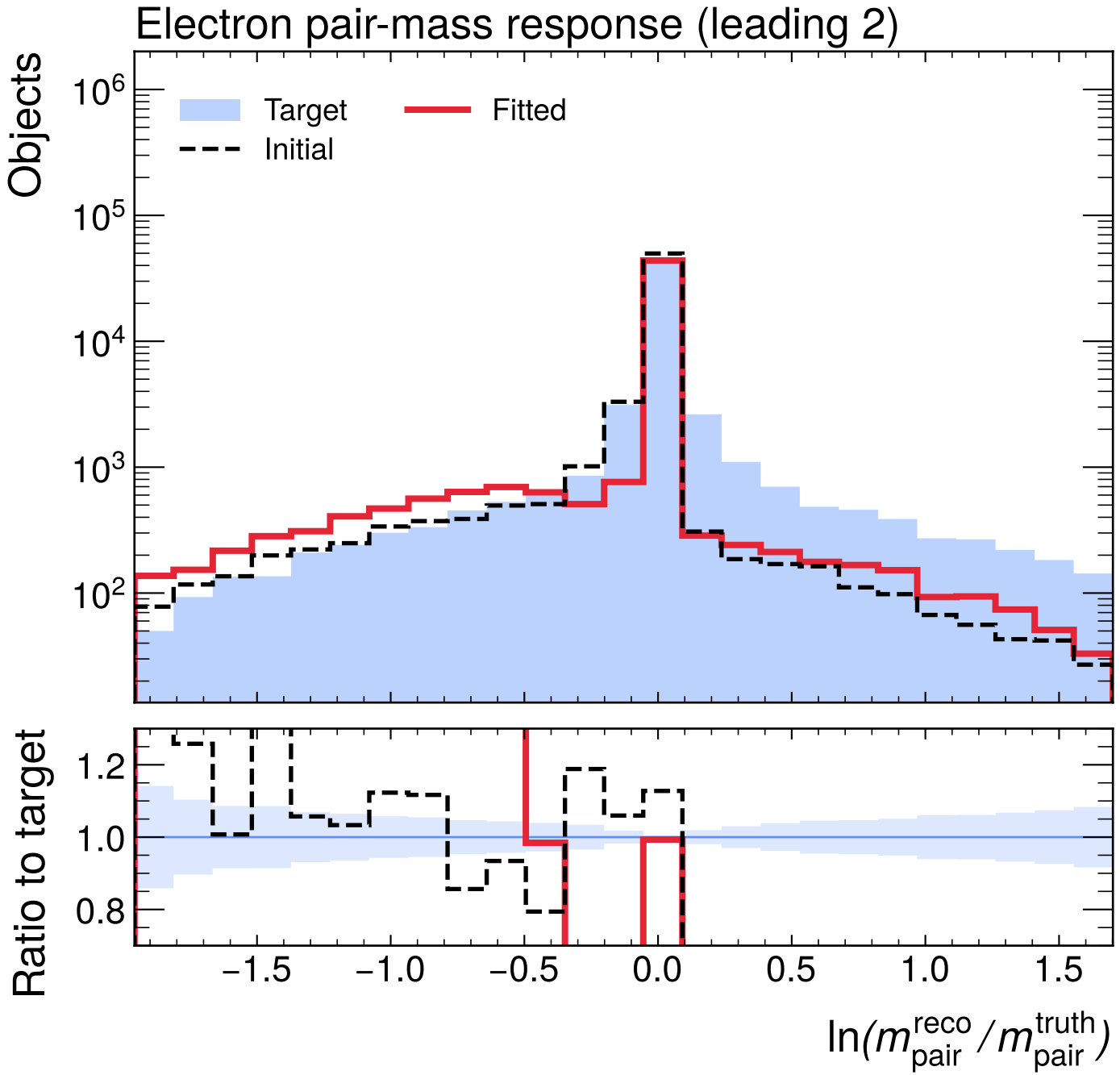

Muon
Objects
Target
Initial
Fitted
10^5
10^4
10^3
10^2
Ratio to target
1.2
1.0
0.8
1
2
3
4
5
6
log(E/GeV)

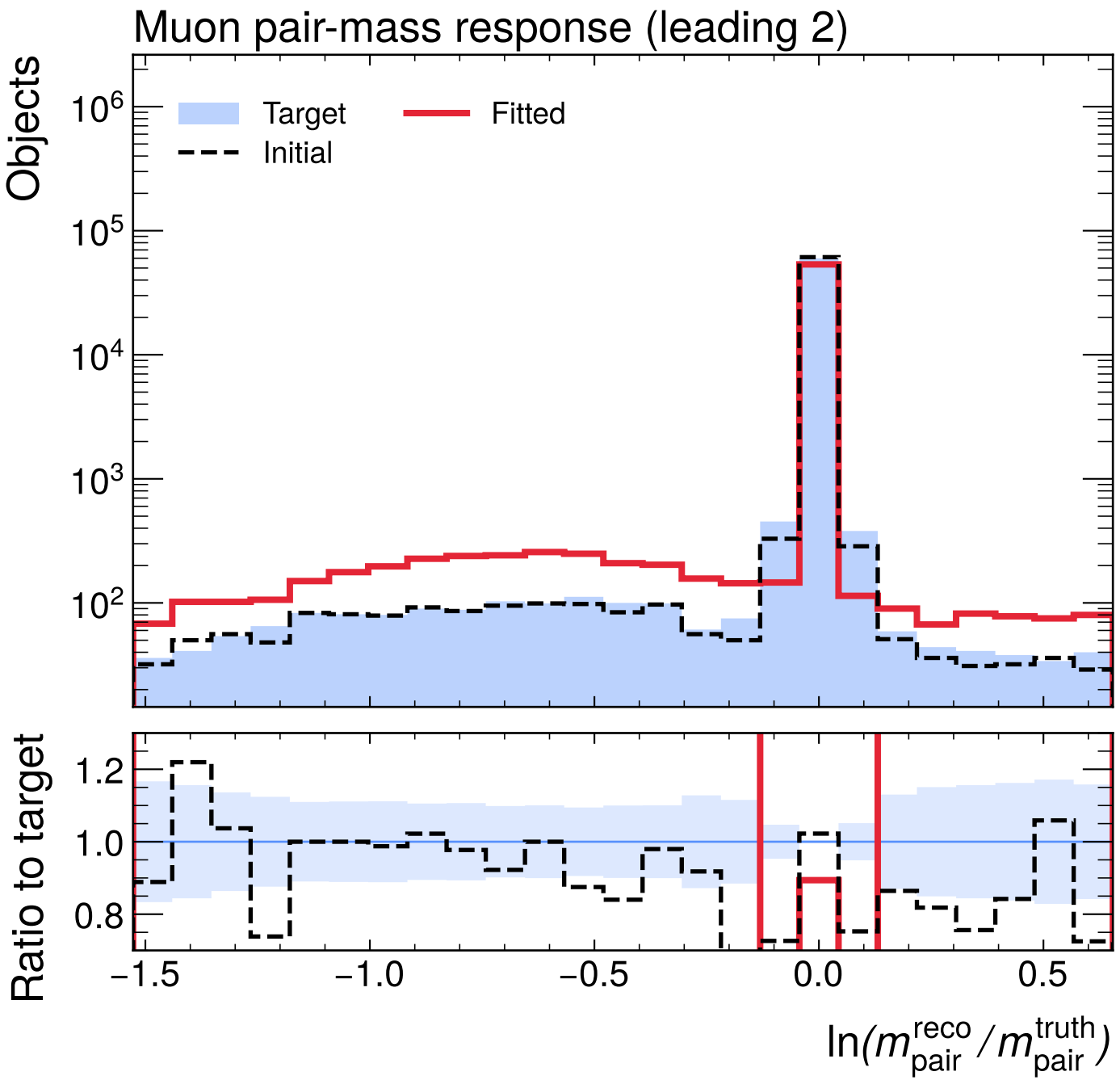

Muon
Objects
Target
Fitted
Initial
2×10^4
10^4
6×10^3
4×10^3
3×10^3
2×10^3
Ratio to target
1.2
1.0
0.8
−2
−1
0
1
2
η

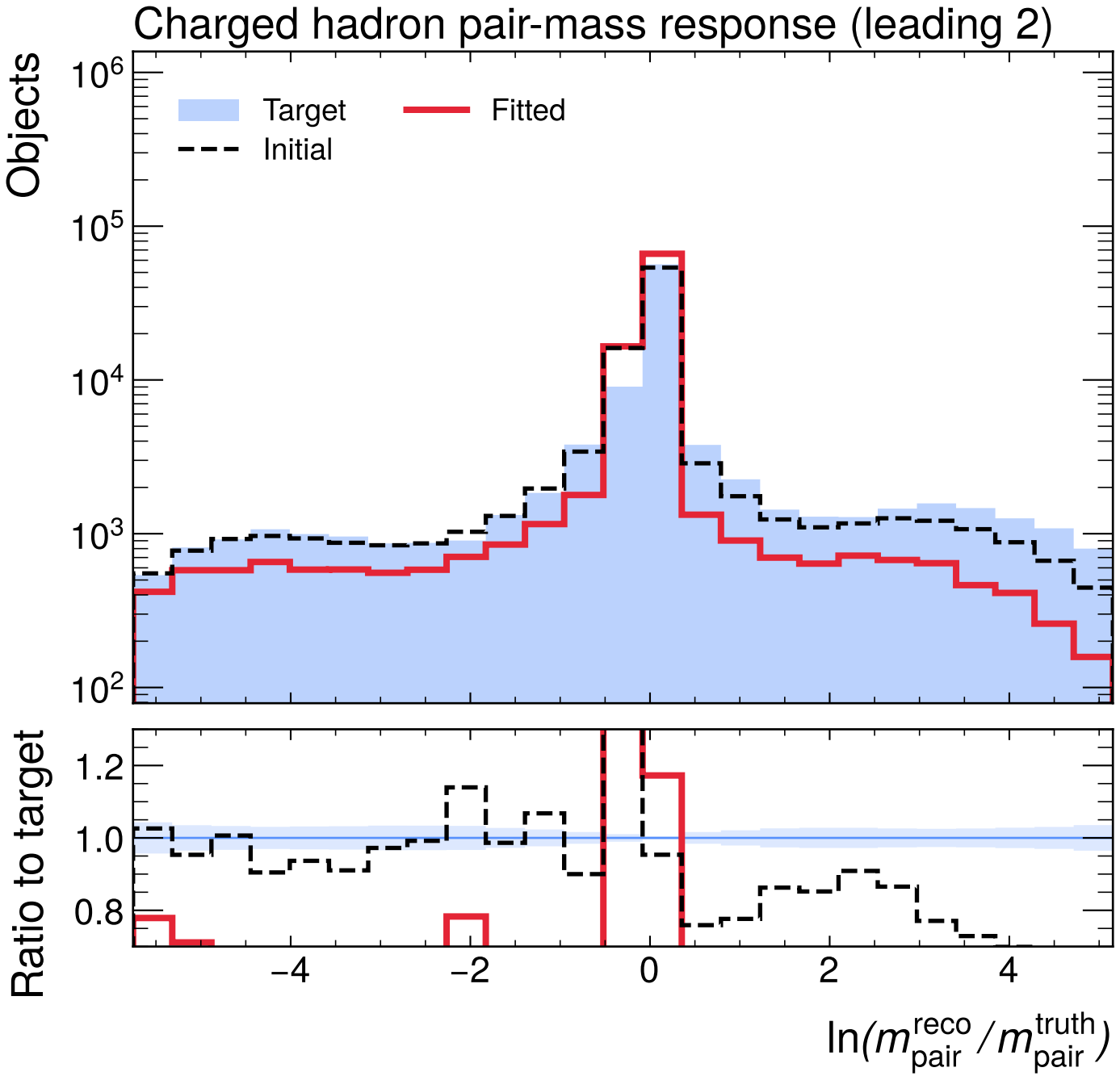

Photon
Objects
Target
Initial
Fitted
10^6
10^5
10^4
10^3
Ratio to target
1.2
1.0
0.8
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
$\log(p_T/\mathrm{GeV})$

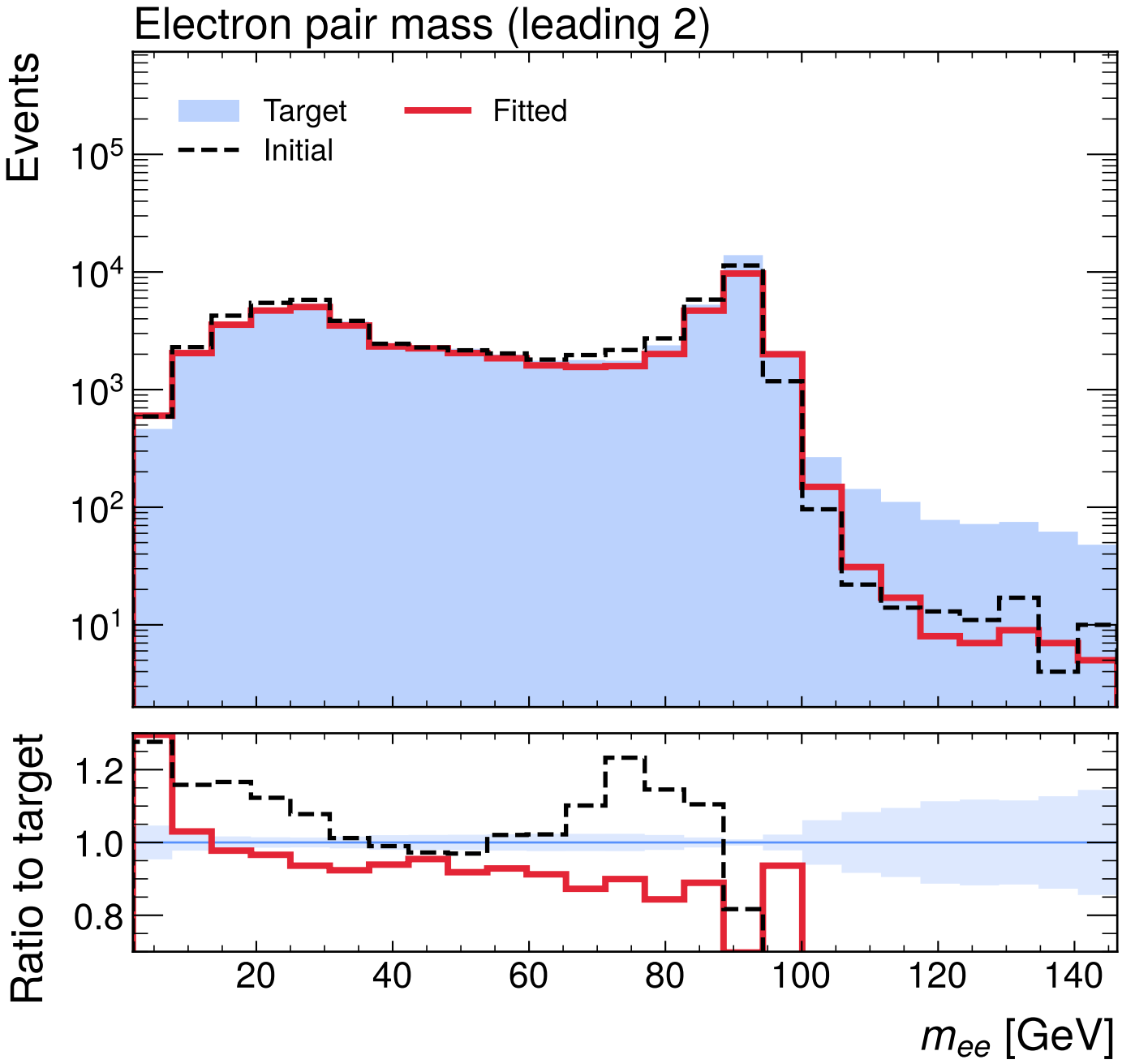

Photon
Objects
Target
Initial
Fitted
10^6
10^5
10^4
10^3
Ratio to target
1.2
1.0
0.8
1
2
3
4
5
log(E/GeV)

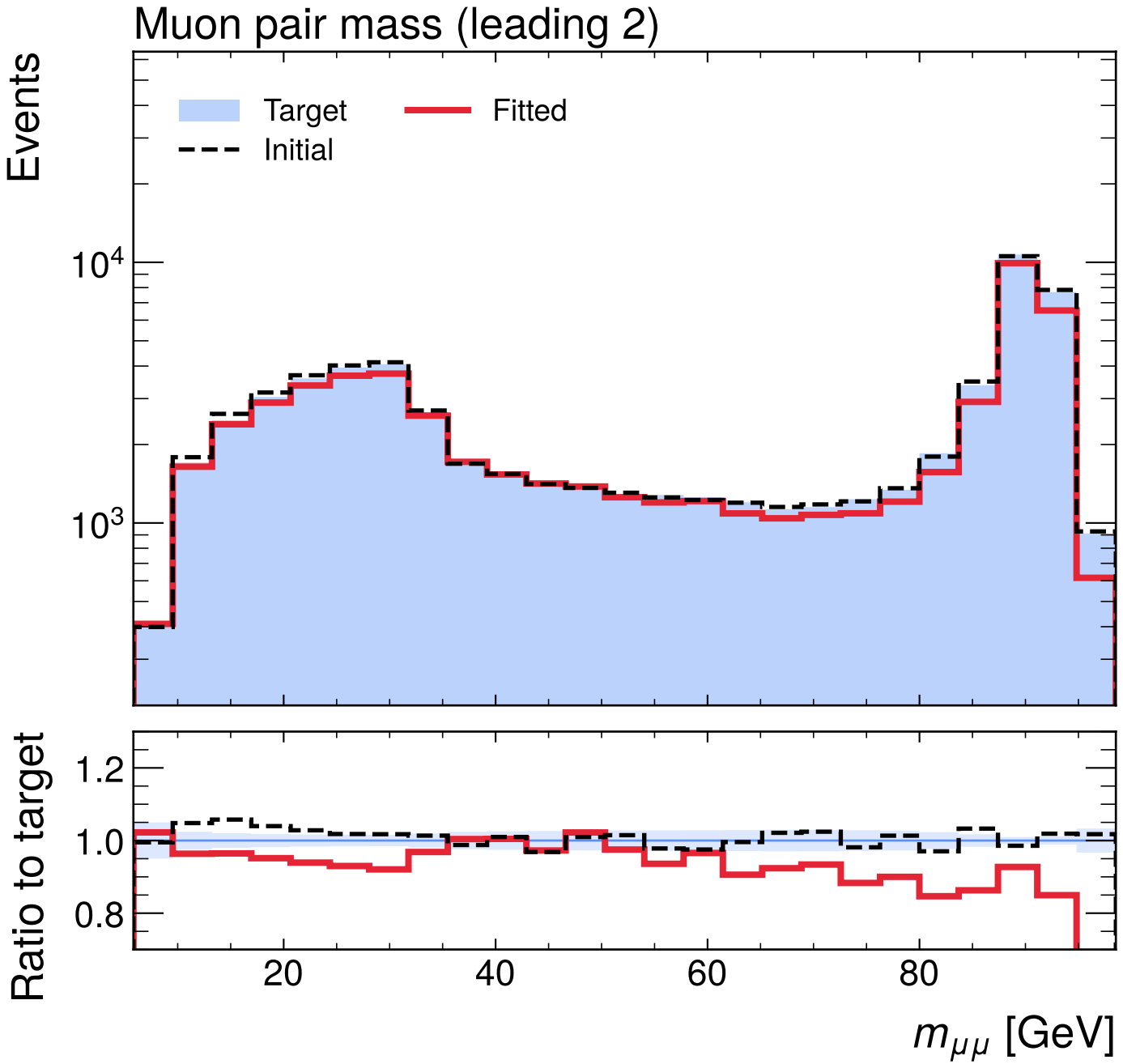

Photon
Target
Fitted
Initial
Objects
10^5
10^4
10^3
10^2
Ratio to target
1.2
1.0
0.8
−2
−1
0
1
2
η

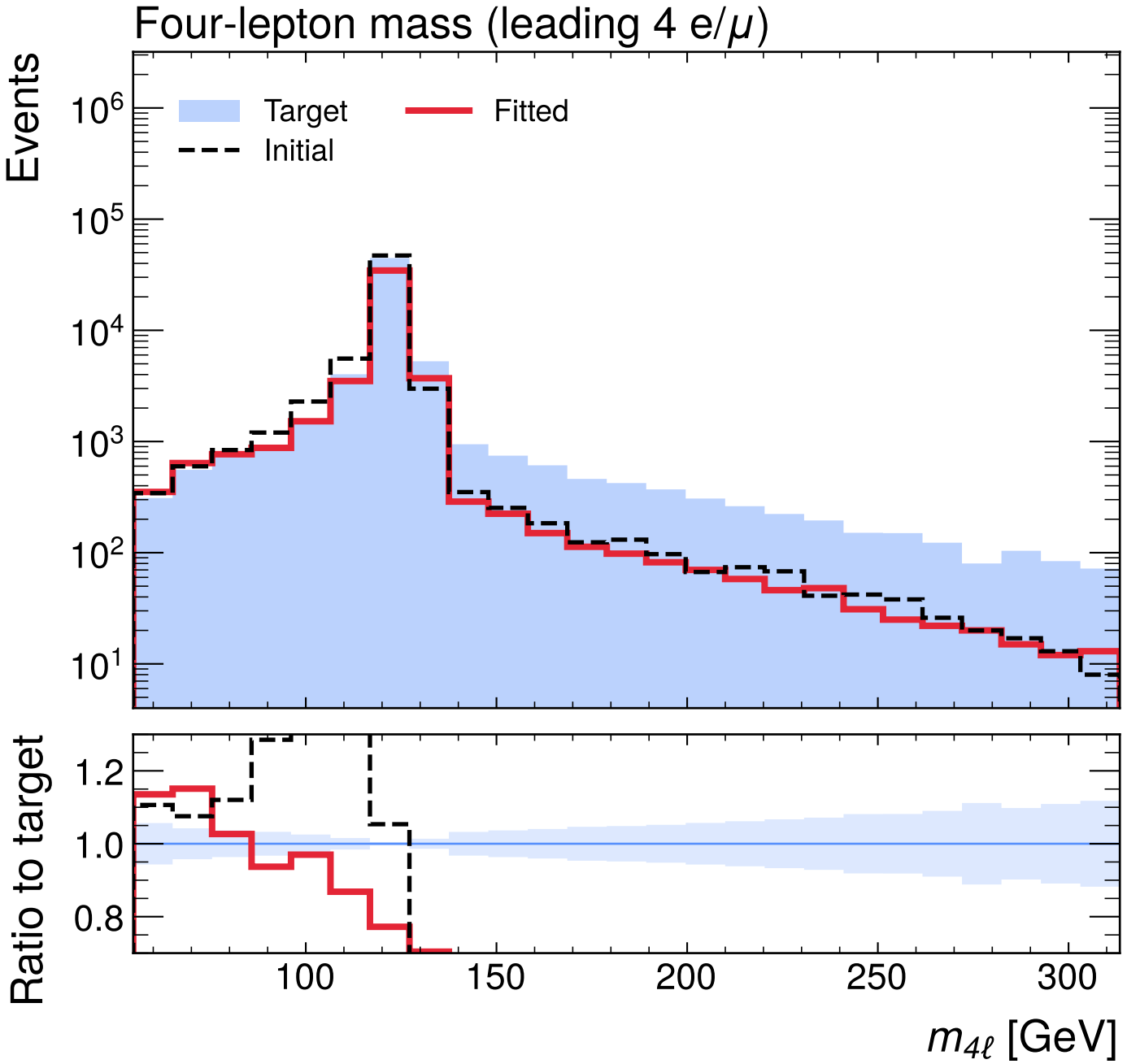

Electron pair-mass response (leading 2)
Objects
Target
Initial
Fitted
10^6
10^5
10^4
10^3
10^2
Ratio to target
1.2
1.0
0.8
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
$\ln(m_{\text{pair}}^{\text{reco}} / m_{\text{pair}}^{\text{truth}})$


Muon pair-mass response (leading 2)
Objects
Target
Initial
Fitted
10^6
10^5
10^4
10^3
10^2
Ratio to target
1.2
1.0
0.8
−1.5
−1.0
−0.5
0.0
0.5
$\ln(m_{\mathrm{pair}}^{\mathrm{reco}} / m_{\mathrm{pair}}^{\mathrm{truth}})$


Charged hadron pair-mass response (leading 2)
Objects
Target
Fitted
Initial
10^6
10^5
10^4
10^3
10^2
Ratio to target
1.2
1.0
0.8
−4
−2
0
2
4
$\ln(m^{\text{reco}}_{\text{pair}}/m^{\text{truth}}_{\text{pair}})$


Electron pair mass (leading 2)
Events
Target
Fitted
Initial
10^5
10^4
10^3
10^2
10^1
Ratio to target
1.2
1.0
0.8
20
40
60
80
100
120
140
m_{ee} [GeV]


Muon pair mass (leading 2)
Events
Target
Fitted
Initial
10^4
10^3
Ratio to target
1.2
1.0
0.8
20
40
60
80
$m_{\mu\mu}$ [GeV]


Four-lepton mass (leading 4 e/μ)
Events
Target
Fitted
Initial
10^6
10^5
10^4
10^3
10^2
10^1
Ratio to target
1.2
1.0
0.8
100
150
200
250
300
$m_{4\ell}$ [GeV]